

REVIEW ARTICLE

A scoping review of engineering education systematic reviews

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Abstract

Background: Systematic review or systematic literature review (SLR) methodologies are a powerful tool for evidence-based decision making. The method originated in the medical sciences but has since been adopted by other disciplines, including engineering education (EE).

Purpose: We aimed to answer two research questions: (i) To what extent is the SLR research method being applied in EE? (ii) How closely are SLRs published in EE following established reporting guidelines for the methodology?

Scope/Method: We searched Inspec, Compendex, and ERIC for engineering-related SLRs and meta-analyses (MAs). We included English language papers that contained an explicit SLR search, or where it appeared the methodology was intended by the author(s). We completed a data extraction process for 21 descriptive and quality-related items, including engineering discipline, which allowed us to identify the EE studies analyzed in this article.

Results: This sub-analysis presents the results of 276 EE-related reviews. We found the use of SLR/MA methods is growing in EE, with 93% of papers published during 2015–2022. However, we found that authors are not generally following established guidelines for reporting their methods and findings.

Conclusions: Not following the best practices for conducting and reporting SLRs can result in the presentation of incorrect summaries and analyses due to missed evidence. Including search experts (e.g., librarians) trained in conducting SLRs can improve review quality. There is also an opportunity for EE-related publishers to recruit experts trained in conducting SLRs as peer reviewers to participate in evaluating submitted reviews.

KEYWORDS

evidence synthesis, knowledge synthesis, librarians, meta-analysis, research methods, systematic reviews

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1 | INTRODUCTION

Systematic literature review (SLR) methodologies are a powerful tool for evidence-based decision making. Although the method originated in the medical sciences, it has since been adapted to address disciplines as disparate as public policy and software engineering. Simply put, an SLR is a formalized application of the literature review, treating the familiar process of searching the literature and writing a review paper as an experiment unto itself. To conduct an SLR, researchers develop and follow a predefined protocol to identify relevant documents, select and analyze the documents that meet criteria for inclusion, extract data from these articles, and synthesize that data to answer a specific research question. The idea behind this process is that leveraging the breadth of the available literature in a given subject area via a rigorous protocol has the potential to produce a replicable and transparent result.

Despite how straightforward this all may sound when framed as an extension of the literature review that every academic should know how to conduct, a properly constructed and performed SLR requires sophisticated search strategies and rigorous adherence to analytical protocols. A poorly conducted SLR with inadequate searching and inconsistent analysis is just as much bad science as a poorly conducted experiment in the laboratory. Several of the authors of this article, being librarians who work with engineering researchers, have anecdotally observed that researchers often fail to fully grasp the complexities of the method. A literature review is not an SLR just because the author is, colloquially, “systematic” in their approach.

Although there is a need for an examination of SLRs in engineering disciplines more generally, engineering education (EE) deserves special attention. Since the 2014 article by Borrego et al. in the *Journal of Engineering Education* (JEE), SLRs have been a hot topic in the discipline. This is further evidenced by a special issue of the JEE dedicated to SLRs and meta-analyses (MAs), to be published in 2023 (Adesope, 2021). Researchers are clearly using these methods to answer questions in the EE space, but the scope of the phenomenon, the level of quality, and the reporting of the methods involved have not been formally studied. This is the gap this study intends to fill.

With all of this as context, we conducted a scoping review (a variant of the SLR that intends to map a phenomenon in a given area of the literature) to determine the prevalence and quality of the methodology of the SLRs in publication in engineering disciplines. It should be noted that this scoping review was intended to cover the breadth of engineering disciplines, beyond just EE. This article, however, presents the subset of the review's results that deals exclusively with the use of SLR methodologies in EE. For the purposes of this sub-analysis, the research questions were (i) to what extent is the systematic literature review (SLR) research method being applied in EE and (ii) how closely are systematic literature reviews (SLRs) that are published in EE following established reporting guidelines for the methodology.

2 | POSITIONALITY STATEMENT

We are four librarians, employed at research-intensive institutions, R1, in the United States of America, who have all previously participated in evidence syntheses projects. We decided to conduct this review based on our experiences working with researchers who are interested in performing SLRs but who often do not have a full understanding of what the methodology involves. We leveraged these experiences in the design of this scoping review, with a focus on the reporting of the methodology authors included in their self-identified SLRs and/or MAs. Three of us have extensive experience collaborating with engineering researchers in teaching and research and were able to apply that experience in this project. One of us comes from the health sciences but brings significant experience in teaching graduate students about SLR methods. All four authors support and strive to elevate transparent and reproducible evidence synthesis practices.

3 | LITERATURE REVIEW

3.1 | History of systematic reviews

The origin of what we now call SLR is somewhat nebulous, as it is the aggregation of literature to provide evidence in support of a hypothesis, which is a natural outgrowth of literature-based research more generally. It is commonly accepted that the first formal SLR in medicine dates back to 1753 in the form of Dr. James Lind's investigation into

methods for treating scurvy (Milne, 2012). A former surgeons' mate in Britain's Royal Navy, Lind conducted a survey of the literature on the scurvy treatments, citing 45 books as meeting his inclusion criteria (Lind, 1757).

Although we can track the development of the SLR to the 18th century, the method's modern form is heavily indebted to work done in the 1990s by the Cochrane Collaboration (Shah & Chung, 2009) and their contemporaries. Cochrane's focus was—and is—primarily on the medical sciences, using SLR methods to promote evidence-based medicine. Intending to bring Cochrane's methods to bear on nonclinical topics, the Evidence for Policy and Practice Information and Coordinating (EPPI Centre) at the University of London promoted SLR methods in areas related to public health and education (Oakley et al., 2005). The Campbell Collaboration also leveraged evidence synthesis methods to investigate the social sciences, with a focus on issues impacting public policy (Littell & White, 2018). The Quality of Reporting of Meta-Analyses (QUOROM) Statement took a discipline-agnostic approach, with a focus on MAs in general (Moher et al., 1999). QUORUM would later develop into the Preferred Reporting Items of Systematic Reviews and Meta-Analyses (PRISMA) statement (Page et al., 2021).

In modern practice, there are a variety of competing methodologies for conducting and reporting SLRs, depending on the discipline being studied. Although PRISMA is a prominent, multidisciplinary tool for reporting an SLR, other guidelines exist in the literature without institutional backing, often associated with particular disciplines with the intent of addressing issues specific to those disciplines. To list some examples, Tranfield et al. (2003) comes out of organizational and management research; Kitchenham (2004) is used to guide software engineering research; Rooney et al. (2014) is based in environmental health; and Mariano et al. (2017) addresses bioinformatics.

With the rise in popularity of SLR/MA came the need to establish a way of ensuring that the same methodological rigor was being applied to SLRs/MAs as had been applied to the primary studies they evaluated. In 2011, the Centre for Reviews and Dissemination created PROSPERO to serve as an open repository for systematic review protocols; however, only reviews related to human health are eligible for registration. In their article announcing the creation of PROSPERO, the PLoS Medicine Editors (2011) likened the need for SLR/MA protocols to the requirement for clinical trials to be registered, namely the need for transparency of the intended goals and methods a research team intends to apply to answer their research question. Although the components required for protocols can vary, common elements include the research question, eligibility criteria, databases to be searched, data to be extracted, methods for data analysis, and quality assessment of included articles. The Cochrane Collaboration publishes the protocols for their sponsored reviews. An alternative to PROSPERO is the Open Science Framework (OSF), which is a general repository but provides timestamps of when documents are uploaded and the ability to provide reviewers with viewing links for private projects. Education-related reviews can look at registering with the Campbell Collaboration, though, like Cochrane, Campbell requires an internal review and approval process to be included. For researchers wanting to have their protocols formally peer-reviewed and published, the journal *Systematic Reviews* accepts full protocol papers, although they also encourage registering the protocol with PROSPERO or OSF (*Systematic Reviews*, n.d.).

3.2 | Systematic literature reviews in engineering

Although most engineering disciplines lack a specific guiding document for SLRs, the incidence of engineering studies identifying themselves as such has increased dramatically since the turn of the century (Reed et al., 2020). The reason for this overall increase is unclear, but it could be attributed to the increased prominence of SLRs in other adjacent disciplines (e.g., medicine, management; Clarke & Chalmers, 2018; Denyer & Tranfield, 2009) or to advocacy within specific engineering disciplines (e.g., EE or software engineering) (Borrego et al., 2014; Kitchenham et al., 2009).

The rise of SLRs in EE is worth examining in more detail, as the tipping point is identifiable. The 2014 article in *JEE* by Borrego et al. (2014) not only highlighted the potential benefits of SLR methods for EE research but framed it specifically within EE's nature as a highly interdisciplinary field.

3.3 | Library support for systematic literature reviews

Given the need for sophisticated and highly technical search strategies, access to and familiarity with research databases, and experience with citation management systems when attempting to follow an SLR protocol, library and librarian support is vital to the endeavor (Aamodt et al., 2019; Meert et al., 2016; Rethlefsen et al., 2015; Schellinger et al., 2021). This support can take several paths, via libraries' traditional roles as supporters and facilitators of

scholarship and as educators (Spencer & Eldredge, 2018). In an academic library environment, this means, respectively, collaborating with scholars to help them better meet the requirements of an SLR, and teaching future scholars about SLR methodologies to better prepare them to conduct their own reviews.

Several universities offer SLR services, in addition to individual librarians working as collaborators and/or co-authors on SLRs. These programs vary from institution to institution, but the connective tissue is based on the consultation model. At the institution three of us work at, Purdue University, librarians often co-author SLRs, and offer orientations to the SLR process and consultations on search strategies for projects that do not have a librarian co-author. At the University of Minnesota, university librarians provide those same services, but also help with developing, registering, and implementing specific SLR protocols (University of Minnesota Libraries, 2021). At the University of Kansas, the service is advertised as a way of adding a librarian to an SLR team, although co-authorship is not specified (University of Kansas Libraries, 2021).

Libraries have developed a variety of instructional activities around SLR methodologies, aimed at both researchers who will be conducting these reviews and librarians and staff who will be supporting these researchers. Librarians at the University of Alberta offer monthly 3-h workshops on systematic review searching (Campbell et al., 2016). The University of Michigan uses a flipped classroom model to deliver a course that prepares librarians to serve on SLR teams (Conte et al., 2015). Purdue University librarians have created a 1-credit graduate-level course on SLR methods (McGowan et al., 2021). The National Institutes of Health Library offers a dozen classes related to different facets of SLR methodologies (NIH Library, n.d.).

4 | METHODS

The EE papers in this study are a subset of a larger data collection effort. For this reason, we first describe how the larger dataset was collected and subsequently narrowed to the relevant EE-related studies.

4.1 | Protocol

This study followed an a priori protocol based on the PRISMA extension for Scoping Reviews (PRISMA-ScR; Tricco et al., 2018) and the article by Arksey and O'Malley (2005) on scoping review best practices. This protocol has been registered with the OSF with documentation available at <https://osf.io/de3mq/>. We chose OSF because it provides date stamps for uploaded and updated materials, and because it is an open repository providing the same level of transparency and accessibility for our scoping review protocol as for registered SLR protocols. A preliminary report on this project, with methodological details, was previously presented at the Frontiers in Education Conference (Reed et al., 2020).

4.2 | Eligibility criteria

4.2.1 | Inclusion criteria

To qualify for inclusion in this study, documents must have satisfied the following conditions:

1. The documents must be journal articles or conference papers. We chose to include conference papers because it is common practice in engineering disciplines (particularly in EE) to present peer-reviewed work at conferences.
2. The full text must be in English. Given our language skills, we are not able to fully assess documents in other languages.
3. The study type must be an SLR and/or MA that includes an SLR-style search, or the authors intended to follow one of those methodologies. We excluded studies that claimed to be SLRs or MAs but shared little or no details (e.g., database selection, search terms) on their data collection methods.
4. The document must include a focus on engineering, excluding software engineering and computer science. We define "a focus on engineering" as (i) an author with an engineering affiliation, or (ii) publication in a journal or conference in the engineering field, or (iii) the application of engineering expertise in non-engineering subject areas (e.g., medicine), or (iv) for EE studies, if engineering or STEM or STEAM appeared in the article title. For studies with STEM or STEAM in the article title, studies were included only if there was an EE component.

4.2.2 | Exclusion criteria

1. Studies that were not journal articles or conference papers were excluded, such as book chapters or dissertations. We chose not to include other document types because they are not typically used to disseminate SLRs and MAs.
2. Studies where the full text was not available in English were excluded.
3. Studies that did not identify as a systematic review, SLR, or MA in the title or abstract were excluded. Studies were also excluded that claimed to be SLRs or MAs but did not report anything in their methods that allowed us to extract data.
4. Studies that were identified as works in progress or preprints were excluded. As these works had not completed the publishing process, we believed there was potential the final methods reported might change based on completing the project and going through peer review.
5. Studies that did not include a focus on engineering were excluded, as were studies that were on software engineering or computer science. We chose not to include software engineering and computer science because they have a distinct and long-standing set of disciplinary guidelines for SLRs already in place (Kitchenham, 2004). We did include EE-related studies that had aspects of software engineering (e.g., software engineering students). For articles with STEM or STEAM in the article title, studies were excluded if the study did not include EE, for example, STEM in agriculture education.

4.3 | Information sources and search

An engineering librarian with experience in conducting systematic reviews (MP) designed the initial search strategy, which was then reviewed by two additional librarians—one an engineering librarian (AvE) and the other a health sciences librarian with review project expertise (JBR). The databases searched were Compendex (using the Engineering Village platform, coverage from 1884) and Inspec (using the Engineering Village platform, coverage from 1896). We chose these databases because of our focus on the engineering literature and their broad coverage of engineering subject areas, with the recognition that we may have missed engineering-focused studies that would have been found in a more dedicated disciplinary database (e.g., SciFinder-n for chemical engineering). The search string (see Figures 1 and 2) combines natural language searching for the phrases (“systematic review” OR “systematic literature review” OR “meta-analysis” or “meta analysis”) in the subject/title/abstract field with terms for engineering disciplines derived from the National Academy of Engineering subject list (National Academy of Engineering, n.d.), using the Engineering Village platform's syntax with no added filters. Preprints were specifically excluded to make assessment of inclusion criterion no. 1 easier to manage. The search was initially conducted on July 25, 2019. The search was run again on October 24, 2022. This search update also added the ERIC database (using the EBSCOhost platform, coverage from 1966), with the initial search adapted for this new database by JBR, to better capture EE content, specifically.

4.4 | Selection of sources of evidence

4.4.1 | Title and abstract review

The title and abstract review was completed using the open-source Rayyan software (Ouzzani et al., 2016) for the 2019 dataset and Covidence for the 2022 expanded dataset (Covidence, 2023). In both cases, the approach was overly

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(((((("systematic review" or "systematic literature review" or "meta-analysis" or "meta analysis")
WN KY) AND (("aerospace engineering" OR "aeronautical engineering" OR "astronautical
engineering" OR "bio* engineering" or "biomedical engineering" OR "chemical engineering"
OR "civil engineering" OR "construction engineering" or "environmental engineering" or
"structural engineering" or "electronic engineering" OR "electrical engineering" OR "computer
engineering" OR "energy engineering" OR "nuclear engineering" OR "industrial engineering" or
"manufacturing engineering" or "systems engineering" OR "operations research" OR "materials
engineering" OR "mechanical engineering" OR "engineering education") WN ALL)))) NOT
({pp} WN DT))
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FIGURE 1 Full search strategy for Compendex and Inspec.

1. DE "meta-analysis" OR DE "State of the Art Reviews"
2. TI ("systematic review" or "systematic literature review" or "meta-analysis" or "meta analysis") OR AB ("systematic review" or "systematic literature review" or "meta-analysis" or "meta analysis") OR KW ("systematic review" or "systematic literature review" or "meta-analysis" or "meta analysis")
3. S1 OR S2
4. DE "Engineering Education" OR DE "Aerospace Education" OR DE "Engineering" OR DE "Agricultural Engineering" OR DE "Chemical Engineering" OR DE "Civil Engineering" OR DE "Human Factors Engineering"
5. TI engineering OR AB engineering OR KW engineering
6. S4 OR S5
7. S3 AND S6

FIGURE 2 Full search strategy for ERIC. [Correction added on 13 November 2023, after first online publication: Removed dashes (-) in between “meta analysis” in two instances in the 2nd numbered list of Figure 2.]

inclusive: studies were excluded if and only if they clearly did not meet the inclusion criteria. Uncertain cases were deferred into the full-text review.

The inclusion criteria were piloted before the 2019 title and abstract review, using the blinding feature in Rayyan. Each of the three reviewers reviewed the same 100 abstracts and tracked include/exclude decisions. The agreement percentage was 88.7%, with a Fleiss kappa inter-rater reliability percentage of 43.7% (moderate agreement), and a meeting was held to reach consensus on the discrepancies. The remaining abstracts were then each reviewed by a single reviewer. This process was repeated for the 2022 dataset, albeit with four reviewers instead of three and no pilot phase.

4.4.2 | Full-text review

The full-text reviews were also completed using Rayyan in 2019 and Covidence in 2022. Before the beginning of the full-text review process in 2019, a fourth reviewer (DZ) was added, who is also an engineering librarian with systematic review experience. There was another pilot test, this time consisting of two rounds of 44 documents apiece. Each set of 44 was randomly selected from the overall set of full-text review records. These pilots were performed by all four reviewers to ensure the consistency of the final dataset. At the completion of pilot testing, we determined that for this stage dual independent review was needed, and so all the remaining articles were reviewed for inclusion by a minimum of two reviewers. In cases of disagreement, the two reviewers discussed the record, and if a consensus could not be reached, that record was reviewed by the entire team. One of the issues that arose in determining whether a study should be included was the prevalence of articles that included one of the phrases “meta-analysis,” “systematic review,” or “systematic literature review” in the title, abstract, or subject yet upon full reading did not report methods that matched those study types. Given the large number of results, we decided not to contact specific authors, but instead, if the methods section included at least one of elements we were looking for in the data extraction, we took the authors word that the study was an SLR or MA. This may have resulted in including articles where the authors had not intended to perform one of these study types and/or we may have excluded studies that were intended to be those methodologies yet did not describe their methods in a way that allowed us to extract data. This process was repeated in 2022, but the change to Covidence meant that each document could be assessed by two reviewers, with a third reviewer intervening in cases of disagreement.

4.5 | Data charting process

Data was extracted using a custom spreadsheet in Google Sheets. The specific items extracted were based on components of the 2009 PRISMA checklist (Moher et al., 2009), as well as the AMSTAR (A MeaSurement Tool to Assess systematic Reviews; Shea et al., 2017) and DART (Documentation and Appraisal Review Tool; Diekemper et al., 2015) for assessing SLR quality. The specific data components are explained below, but largely focused on article demographic data, components of SLRs/MAs that are described in the methods, such as databases searched and search strategy, and

components reported in the results, such as a flow diagram of the search screening process. A pilot test was also conducted for this phase of the review, with all four reviewers extracting data for the same 18 articles. This test and the subsequent discussions resulted in clarifications to several of the data extraction fields; the final extraction template can be found at <https://osf.io/de3mq/>. The remaining documents were then divided into four groups, one for each reviewer, and extracted. When the extracting reviewer was unsure of how to answer a question, it was presented to the group as a whole and a consensus was reached.

4.6 | Data items

- Bibliographic data:
 - Basic article information: For each included article we extracted, the last name for the first author, the year the article was published, the country of the corresponding author (or the first author if not reported), the source title (journal or conference proceeding where the article was published), and the publication title (title of the article or conference paper).
 - Engineering discipline: The list of engineering disciplines was the same list, based on the National Academy of Engineering subject list, used in creating the search strategy and in the criteria for inclusion in the final dataset. This data element did involve a level of subjectivity that required a judgment call because of the amount of multi-disciplinary work performed within engineering research. When multiple disciplines were represented in an individual article, we checked the journal or conference title to see if a specific discipline was present and checked the first or corresponding author's affiliation. If a specific engineering discipline was not listed in these sources, we went down the author list and selected the first discipline listed in an author affiliation.
 - Systematic literature review or meta-analysis in title: The next two data extraction elements were whether the final article self-reported as an SLR or systematic review and/or MA in the title and if so which one. A drop-down option for yes or no was used for reporting in the title. For recording which variation, if any, was reported, the drop-down options were SLR, MA, both, neither.
- Review protocol data:
 - Clearly defined research question: For this category, we looked for an explicit research question, often located at the end of the introduction. To be marked as included, the authors needed to explicitly state it as the research question, variations like Q1, or in question format. Studies that reported the objective of the study as performing an SLR of the literature on the topic of the paper were marked as no for this criterion. We made this decision as one of the purposes of an SLR is to answer a specific research question.
 - Registered protocol: Registering an SLR protocol promotes transparency and accountability for sticking to the a priori nature of conducting an SLR or MA. Although many SLRs in engineering would not be eligible for registration with PROSPERO—a registry used for health sciences systematic reviews—researchers could register protocols with repositories such as OSF. Authors had to explicitly state where their protocol was registered or provide information on how to access their protocol.
 - Explicit statement of inclusion/exclusion criteria: To be marked as yes for this component, authors had to list the predetermined criteria that they used to determine the eligibility of discovered studies in the final SLR/MA. Criteria could be presented either as a figure or described in text but needed to be explicitly stated as being inclusion/exclusion criteria.
 - Explicit statement that review methods were established before data extraction: In order to be marked as yes, authors needed to state that the review methods were established prior to beginning the data extraction. This follows the established practices of PROSPERO requiring protocols to be submitted prior to data extraction. This requirement exists to ensure that SLR/MA researchers do not change the data they plan to extract based on the results of the studies they find. Any study that reported uploading a protocol was also marked as yes for this criterion.
 - PRISMA flowchart or list of excluded studies and reasons why: The PRISMA flowchart is a well-known and often-used figure that shows the progress of studies in SLRs/MAs from the initial search to final inclusion (Page et al., 2021). The purpose of this flowchart is to provide transparency on how the review criteria was applied in the study. Any study that included either the official PRISMA flowchart or their own flow diagram showing the number of studies included and excluded at each stage of the review was marked as yes for this category. Additionally, if study authors described in text the flow of articles in each step of the review and why articles were excluded, then those articles were marked yes as well.

- Explicit statement of assessing risk of bias of included studies: To receive a yes for this criterion, studies needed to either explicitly state that articles included in the SLR/MA were assessed for quality or risk of bias or the authors needed to include the name of the tool used to assess the quality of each included study.
 - Explicit statement of multiple reviewers or explanation of inter-rater reliability: In order to be classified as yes for this criterion, studies needed to explicitly state whether each record was reviewed by more than one reviewer, or they needed to describe their process for piloting the process and/or provide an inter-rater reliability measure. Although inter-rater reliability is not a perfect measurement, it demonstrates that the authors of the study were aware of the importance of verifying that different authors might apply criteria differently.
- Search data:
 - Date search run: In order to be marked as yes for this category, articles needed to provide the month and year that the searches were run. Providing this information is important for the reproducibility of SLRs, as it provides additional context for what articles the researchers could have discovered with their search. We did not require the reporting of the specific day, as many engineering-related databases do not provide a search filter option for a specific day of indexing in the database.
 - At least one full database search string provided: In order to be marked as yes for this category, authors needed to provide either a figure of the search taken straight from one of the databases, or provide enough detail in the written description to allow a reader to replicate the search in the reported database. This would include the specific keywords used, the search fields used, and how the terms were combined.
 - Database/s search strategy provided (general themes or concepts): To be marked as yes in this category, authors needed to provide a list of terms or concepts they used in their search for articles, but not necessarily the level of detail needed to reproduce the search. All studies marked as yes for reporting at least one full database search string were also marked yes for this category.
 - Search filters identified: To be marked yes in this category, authors had to report any limiters or search filters applied to their search; this could be reported in either the full search string or as text detailing the search process. Studies were also marked yes if they explicitly stated no search filters or limits were applied. Examples of search filters that might be available and used for SLR or MA include publication date, language of article, and publication type (article, conference paper, abstract, etc.).
 - Search quality: Studies were marked as high or low search quality based on all the factors related to the description of the search strategy. Studies could receive a yes for at least one full database search string provided, but receive a low quality if the reported search was limited in scope, a single source was searched, or only a couple of keywords were used. Each reviewer drew on their experience as database experts to make this determination.
- Sources data:
 - List of sources: This was a yes/no category measuring whether the study authors reported where they searched for articles for their study. Sources could include databases, platforms (such as EBSCO or specific publishers like Wiley), repositories, and/or specific journals searched.
 - Database platforms listed: An important component in reporting the search strategy is not just listing the sources searched but also, in the case of databases, which platform the database is hosted on. This is important, as different platforms provide different search options and it can be impossible to replicate a search if the platform is not identified. On the reverse side is the reporting of what databases were searched if platforms are reported. For instance, EBSCO is a common research platform subscribed to by academic libraries; however, each library chooses which databases hosted by EBSCO they want to subscribe to and researchers can choose which of the databases available to them that they want to search in. If a study reports searching EBSCO, that is not sufficient information to determine whether appropriate databases were searched. To be marked as yes for this category, a study had to report the database and platform for each source that is available in multiple platforms.
 - Total number of databases/platforms/repositories searched: This element is a numerical count of the total number of databases/platforms/repositories listed. If authors reported searching a platform that hosts multiple databases, but did not specify which databases they searched, that was counted as one. If Google Scholar was reported in the list of other databases searched, that was included in the count; if however, Google Scholar was mentioned as an additional component of the search, that was not included in this count. This decision was made as some researchers use Google Scholar as part of their strategy for finding unpublished reports or tracking citing/cited articles and not as part of the reproducible search strategy. Specific journals that were searched were also not included in this count.

- List of databases/platforms/repositories searched: Each of the sources that authors reported as being part of their search, excluding specific journals, was listed here to investigate what sources are commonly used in engineering SLRs and MAs.
- Librarian participation data:
 - Librarian or database expert listed as author: Affiliations were checked for librarian or information expert identification.
 - Librarian or database expert consulted: The full text was examined to see whether the study authors reported meeting or discussing their search with a librarian or database expert. If included, this was listed in the methods or acknowledgment section. Any studies marked as yes for librarian or database expert listed as an author were also marked yes here.

4.7 | Data cleaning

After each reviewer completed their data extraction, the completed sheets were combined into a single dataset and one reviewer (MP) cleaned the data. This process included steps such as resolving different methods of describing databases searched (e.g., Web of Science and WoS) and resolving instances where the publication source was a conference proceeding that included the specific year. In these cases, the source listings were edited to reflect the overall conference proceeding to capture all the included reviews that were published through that avenue. Some data cleaning was performed for the country of the corresponding author, including spelling out country abbreviations (e.g., UAE changed to United Arab Emirates). Additionally, the data was cleaned to correct misspellings with publication titles and to apply consistency to the use of “&” and “and” in publications titles (e.g., Computers & Education vs. Computers and Education).

4.8 | Synthesis of the results

All data analyses were performed in Google Sheets. The studies identified during data extraction as “Engineering Education” were filtered from the larger dataset and copied into a new Google Sheet. The extracted studies represent the final dataset we present in this article.

5 | RESULTS

As mentioned previously, this article presents a sub-analysis of data that we gathered for a larger project. This sub-analysis focuses only on articles related to EE. However, to understand how we obtained the articles for our EE sub-analysis, it is necessary for us to explain the overall review process for the larger project. In total, we obtained 8,193 records from our database searches (conducted initially on July 25, 2019 and updated on October 24, 2022) and after completing a title- and abstract-level review, we were left with 5,840 records. Next, we completed a full-text screening and were left with 3,069 studies in the overall review.

During data extraction, we classified each of the 3,069 included studies by a primary discipline, and we identified 276 of the articles as “engineering education.” In this article, we present the data extraction results of the EE papers by sharing summaries of characteristics of the articles, as well as details of how well the studies adhered to the items on our data extraction quality form. We have uploaded a spreadsheet with the citation details of the 276 engineering education articles into our OSF project, which can be accessed at <https://osf.io/de3mq/>. Figure 3 illustrates our overall study identification and screening process.

5.1 | Summary of the engineering education studies

Geographical distribution: Figure 4 shows that there is global interest in SLRs and MAs in the discipline of EE. Authorship of SLRs and MAs is heavily weighted towards the United States of America, with American corresponding or first authors credited with 105 of the included studies ($n = 276$). Following the United States of America

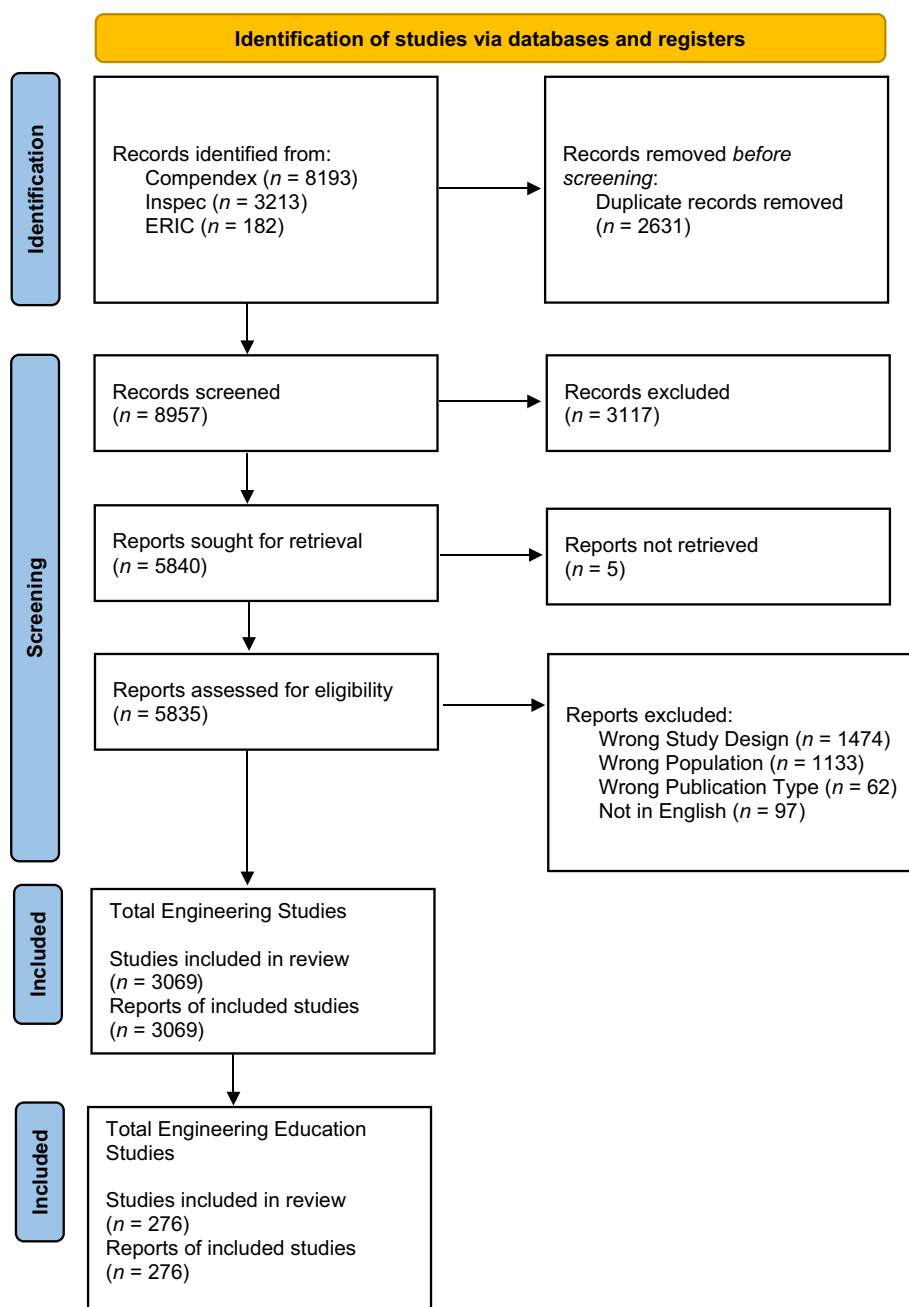


FIGURE 3 PRISMA flow diagram for this scoping review. Source: Adapted from Page et al. (2021).

were Brazil (17), Indonesia (14), China (13), and Malaysia (11). An additional 40 countries accounted for nine or fewer studies apiece. It should be noted that our study was limited to articles written in the English language.

Paper counts by year: Figure 5 shows that conducting SLRs and MAs in EE is a recent practice, as 257 of the 276 studies were published during 2015–2022. Further, 137 studies were published from 2020 to 2022, accounting for 50% of the total studies included. Only 19 studies were published prior to 2015, with the earliest being Springer et al.'s (1999) meta-analysis of STEM undergraduate small group learning studies.

Publication venues: Overall, the EE review papers are published in 113 different venues. Of these, 58 are journals and 55 are conferences. Table 1 shows the top 10 most frequent publication venues for EE-focused SLRs and MAs. Of the EE reviews, 61 were conference papers presented at either the ASEE Annual Conference or the IEEE Frontiers in Education (FIE) Conference.

Databases/repositories searched: Collectively, the authors of the 276 EE SLRs and MAs reported searching 167 unique databases/repositories/platforms. This does not include individual journal titles, which were listed in some

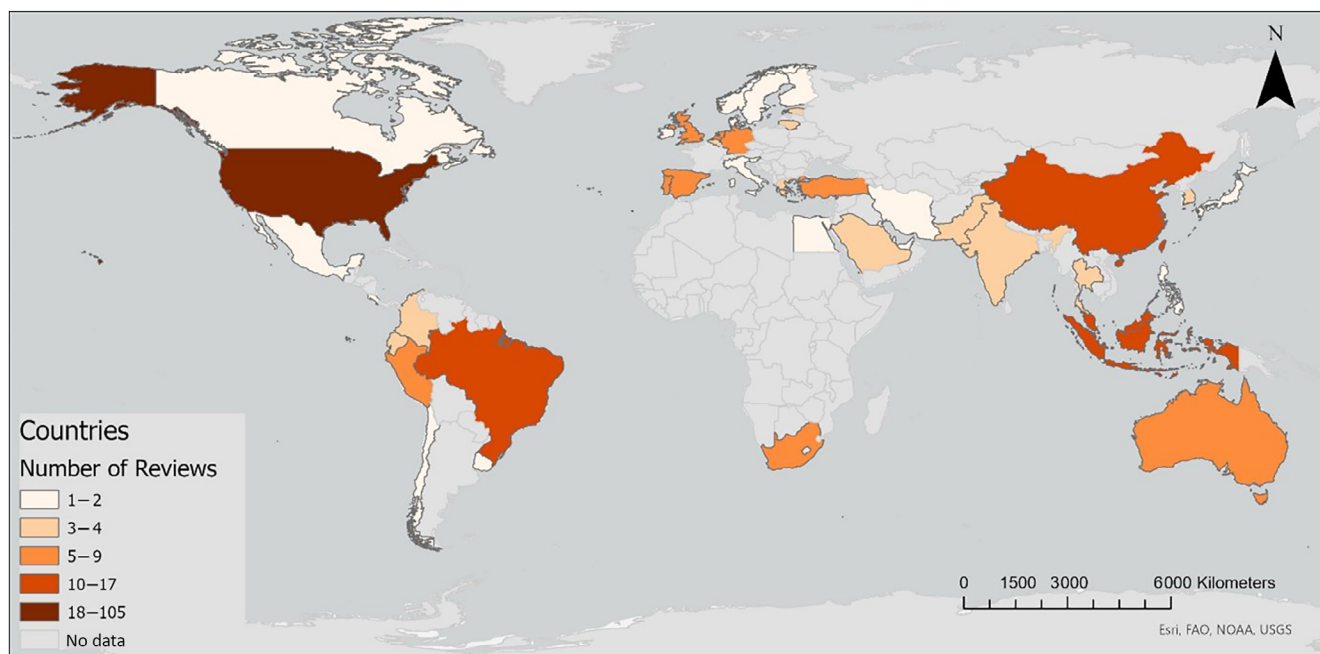


FIGURE 4 Geographic distribution of SLRs and MAs.

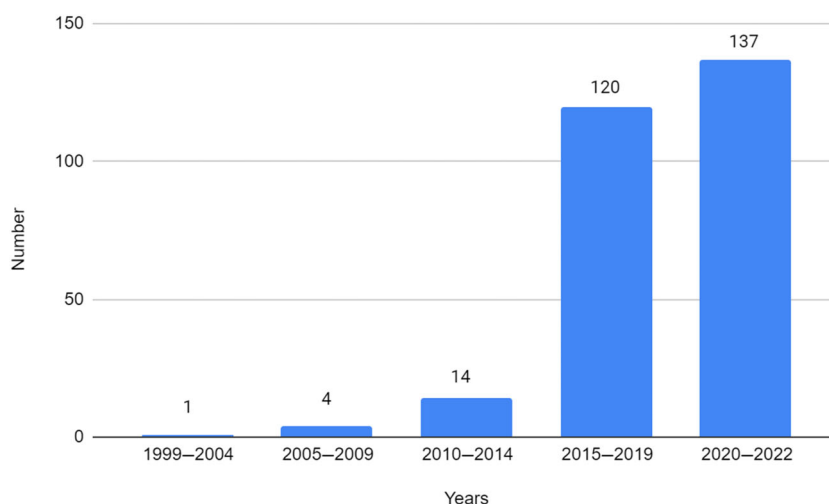


FIGURE 5 Engineering education paper counts by year ($n = 276$).

of the articles. Table 2 details the databases/repositories/platforms that were reported in at least 10 different articles. The most searched database was Scopus, followed by IEEEExplore, Web of Science, and ERIC. Figure 6 shows the number of databases/repositories/platforms searched compared to the number of articles in the dataset that reported searching that many databases/repositories/platforms. Nearly 25% of the SLRs and MAs either did not report any databases/repositories/platforms searched (10) or reported only searching one database/repository/platform (57), which is concerning when SLRs are intended to be exhaustive repeatable searches of the literature. The descriptive statistical values for the number of databases/repositories/platforms searched in the EE SLRs and MAs were as follows: mean = 4, median = 3.5, mode = 1.

Although the authors reported searching 167 unique databases/repositories/platforms, some of the listings are platforms that contain multiple databases (e.g., ProQuest, Web of Science, Engineering Village) and it is not possible to know which databases a given library subscribes to on a platform at a given time. Therefore, like the weaknesses noted in Borrego et al. (2015) study of EE SLRs, as well as Harari et al. (2020) evaluation of applied-psychology-focused SLRs, it is not possible to know the true number of databases searched.

TABLE 1 Most frequent publication venues for engineering education SLRs and MAs.

	Journal/conference	Publisher	Counts
1	ASEE Annual Conference and Exposition Conference Proceedings	ASEE	40
2	IEEE Frontiers in Education Conference (FIE)	IEEE	21
3	<i>Journal of Engineering Education</i>	ASEE (Wiley-Blackwell platform)	17
4	<i>European Journal of Engineering Education</i>	SEFI (Taylor & Francis platform)	15
5	<i>International Journal of Engineering Education</i>	Dublin Institute of Technology-Tempus Publications	12
6	European Society for Engineering Education Annual Conference (SEFI)	SEFI	8
7	IEEE Global Engineering Education Conference (EDUCON)	IEEE	8
8	Computer Applications in Engineering Education	Wiley	7
9	Computers & Education	Elsevier	6
10	IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE)	IEEE	5

TABLE 2 Most frequent databases/repositories/platforms reported as searched by authors of engineering education SLRs and MAs.

Database/repository/platform	Counts
Scopus	106
IEEEExplore	93
Web of Science	92
ERIC	89
ScienceDirect	71
ACM Digital Library	53
Google Scholar	51
Springer	41
Wiley	28
PsycInfo	27
ProQuest	26
Compendex	24
ASEE PEER	20
Engineering Village	20
Taylor & Francis	19
EBSCO	17
Academic Search Complete	16
Education Source	16
Inspec	16
JSTOR	16
ProQuest Dissertation and Theses	13
Sage	12
Academic Search Premier	10

The platform Web of Science can be especially confusing for SLR reporting. As shown in the Clarivate website (Clarivate, 2021), libraries have options for subscribing to various databases on the Web of Science Core Collection and/or the Web of Science platforms. As noted on the website, “Institutional subscriptions to Web of Science vary, both

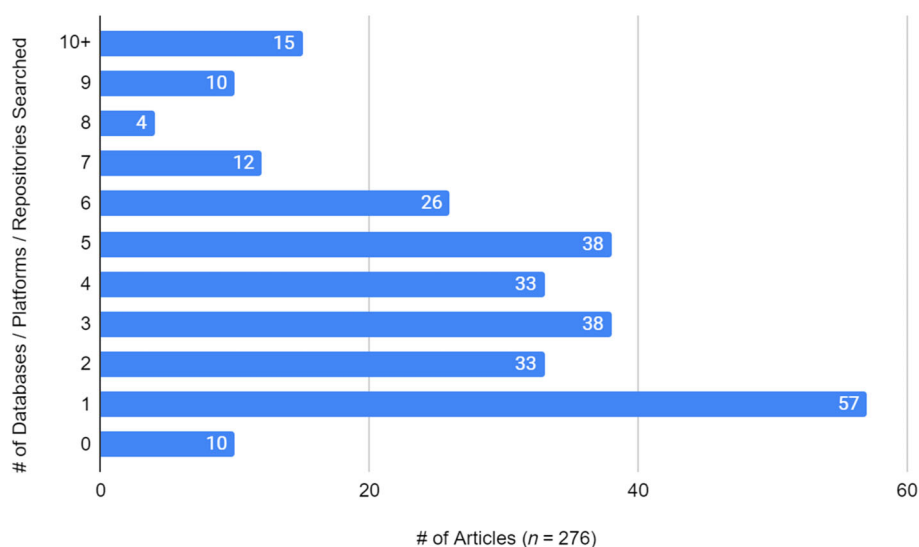


FIGURE 6 Number of databases/repositories/platforms searched compared to the number of engineering education SLRs and MAs reported searching that number of databases/repositories/platforms.

in terms of databases and depth of data. Your institution may have a subset of the content” For this reason, the author of an SLR would need to state the specific database(s) (e.g., Science Citation Index, Conference Proceeding Citation Index) searched.

Additionally, as Harari et al. (2020) discussed, many of the databases/repositories listed were publisher databases (e.g., ScienceDirect, Springer, Wiley, Sage), which contain only literature from individual publishers and have limited search options, as opposed to abstract and indexing databases (e.g., ERIC, Scopus, Compendex), which include content from many different publishers and offer advanced search features. Although it is not harmful to also include publisher databases, searching only these types of databases (or a high percentage) can severely limit the reach of the searches and bias the results toward articles from large publishers. Additionally, it may be repetitive, as the articles from large publishers are typically indexed in disciplinary abstracting and indexing (A&I) databases.

5.2 | Systematic review and meta-analysis quality adherence

In this section, we share the results of our data extraction items related to SLR and MA quality. Table 3 shows the results of 15 data extraction items that had yes/no responses. Overall, the strongest quality performance shown by authors of SLRs and MAs was in reporting a list of sources searched (272 yes and 4 no) and providing at least the general themes or concepts searched (256 yes and 20 no). Although this seems positive, it does not mean the searches are reproducible, which is a key component of SLR reporting. As mentioned earlier, because many of the sources reported being searched were platforms and not individual databases, it is not possible to know which specific databases were searched on a platform. Additionally, it is not possible to reproduce a search when only general themes or concepts are reported since it is not clear how the terms were combined to form the search and whether additional synonyms were used.

The weakest quality performance relates to registering protocols (0 yes and 276 no), including a librarian or database expert as an author (8 yes and 268 no), or consulting with a librarian or database expert (21 yes and 255 no). We believe these issues may be related to a lack of awareness by EE researchers. Some health-science-related journals require protocol registration for publication, but this is not a known practice with EE publishers. Additionally, there are well-known organizations (e.g., PROSPERO) in the health sciences that offer protocol repositories for health-related topics, but this is not the case widely for engineering and other disciplines. Like this scoping review, many authors now use the OSF for protocol registration, but this is a relatively new tool that was originally designed for psychology research reproducibility which has now become multidisciplinary (Foster & Deardorff, 2017). Regarding consulting with or involving librarians in publications, it may be the case that EE researchers are not aware that librarians can provide support for SLR and MA projects.

TABLE 3 Counts of engineering education SLRs and MAs adhering to data extraction quality items by SLR/MA phase ($n = 276$).

Data extraction quality item	No. of papers “yes”	No. of papers “no”
SLR/MA phase: Pre-work		
Clearly defined research question	199	77
Registered protocol	0	276
Librarian or database expert listed as author	8	268
Librarian or database expert consulted	21	255
Explicit statement of inclusion/exclusion criteria	220	56
Explicit statement that review methods were established before data extraction (could be part of registered protocol)	81	195
SLR/MA phase: Searching		
Date search run	81	195
List of sources (databases, platforms, repositories, etc.) searched	272	4
At least one full database search string provided	67	209
Database/s search strategy provided (general themes/concepts)	256	20
Search filters identified	127	149
SLR/MA phases: Screening, data extraction, and assessment		
PRISMA flowchart or list of excluded studies with reasons	133	143
Explicit statement of assessing for bias of included studies	38	238
Explicit statement of multiple reviewers or explanation of interrater reliability	56	220
SLR/MA phase: Publishing		
SLR and/or MA in paper title	199	77

Search quality: In our review of search strategy quality, we classified 31 articles as “high quality” and 245 as “low quality.”

Database platforms listed (for all databases): Of the 276 articles, the authors of 40 publications listed the database platforms for all the databases they named, while 80 did not. For the remaining 156 articles, database platforms were not relevant (i.e., the database Scopus is available only on one platform).

6 | DISCUSSION

Our research questions for this work were to determine (i) to what extent is the SLR research method being applied in EE and (ii) how closely are SLRs that are published in EE following established reporting guidelines for the methodology. Overall, our findings indicate that the use of SLR/MA methods is growing in EE, with 93% of the papers in our results set published in 2015 or later (see Figure 5). This growth may be related to Borrego et al.'s (2014) encouragement of engineering educators to “consider including systematic reviews in their repertoire of methodologies.” However, we found that authors of EE are not generally following established guidelines for reporting their findings for these specialized reviews. In this section we provide insights into our results (from the perspective of librarians), discuss the limitations of our work, and give suggestions for future research in this area.

6.1 | Librarian insights

6.1.1 | Need for continuing education

We feel that there is an opportunity for engineering educators to develop a fuller understanding of the various types of specialized reviews, the current standards/guidelines for conducting and reporting specialized reviews, and the vocabulary surrounding evidence synthesis searching activities.

For evidence synthesis projects, an early consideration that researchers need to make is the type of review that is appropriate for their research question(s) and overall needs. SLRs and MAs are popular and widely discussed approaches, but they are time consuming, taking an average of 67.3 weeks (Borah et al., 2017), and labor intensive, as they are generally conducted in teams with multiple reviewers (Borrego et al., 2015). Additionally, SLRs are designed to address specific research questions, such as those that fit the population-intervention-comparison-outcome (PICO) framework and are not a good fit for overly broad or “what is known about X topic?” questions (Borrego et al., 2014).

A good source to consult when considering review types is Grant and Booth's (2009) article, which provides descriptions, perceived strengths and weaknesses, and example citations for 14 different types of reviews. During this consideration, researchers should examine what review types are most accepted and appropriate for their discipline by reading previously published reviews, exploring author sections of journal websites, and talking with journal editors. For example, scoping reviews, like this one, are conducted to explore what is known about a topic and may be used to determine whether a full SLR is needed (Grant & Booth, 2009). Scoping reviews are quite common in the health sciences but are not yet as common in EE. With regard to EE, a title-abstract-keyword search in the multidisciplinary database Scopus for the key terms “scoping review” and (STEM or engineering) and (education or learning) gives 96 results (as of February 10, 2023), of which 73 results were published from 2021 to 2023. Compared to a similar search, “scoping review” AND (medicine OR “health sciences”) AND (education OR learning) in Scopus returns 593 results, of which 345 were published from 2021 to 2023. This limited example demonstrates that scoping reviews are publishable in EE and, although not as common as in the health sciences, they are growing in popularity, similar to the path systematic reviews have taken. Exploratory searches such as these can also help researchers identify journals and conferences that publish additional review methodology papers.

Once researchers determine the type of review that is best for their work, they should become knowledgeable about applicable and current conducting and reporting standards and guidelines. Engineering educators should start with Borrego et al. (2014) article, which discusses conducting and reporting SLRs in the context of the EE discipline. However, engineering educators should also consult other published resources that go into more depth on conducting and reporting methods for repeatability and transparency. For example, the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement was created initially for health science interventions but has been used widely in other disciplines, including education and social sciences for quantitative and mixed-methods approaches (PRISMA, 2021). The original 2009 PRISMA statement was updated in 2020 to increase transparency and take into account technology changes over time (Page et al., 2021). In addition, PRISMA has created several extensions that are designed to improve the reporting of certain aspects of evidence synthesis projects, including PRISMA-P (PRISMA for Protocols; Moher et al., 2015), PRISMA-ScR (PRISMA for Scoping Reviews; Tricco et al., 2018), and PRISMA-S (PRISMA for Searching; Rethlefsen et al., 2021). The PRISMA-P (Moher et al., 2015) and PRISMA-S (Rethlefsen et al., 2021) extensions are useful for engineering educators conducting any type of evidence synthesis study. Although the protocol extension focuses heavily on PROSPERO, a registry that accepts only health-sciences-related protocols, the information provided about the importance of registration practices and the recommendations for components to include in protocol reporting is helpful guidance to all researchers. PRISMA-P includes both a statement paper (Moher et al., 2015) and an elaboration and explanation paper (Shamseer et al., 2015). The elaboration and explanation paper contains terminology definitions and detailed examples and descriptions that are especially helpful for new evidence synthesis researchers. Likewise, PRISMA-S provides detailed guidance for search strategy reporting, but the searching extension statement and elaboration and explanation details are contained in one paper.

Although the PRISMA statement and its extensions are guidelines for reporting (not conducting) evidence synthesis studies, authors should be familiar with the guidelines well before the reporting stage of a project. This helps the authors to ensure that they are recording enough details about their methods to adhere to the relevant reporting guidelines. In addition to PRISMA resources, it may benefit EE researchers interested in evidence synthesis to explore other options for further education, including readily available materials and in-depth training opportunities through workshops and classes. The *Handbook of Research Synthesis and Analysis* (Cooper et al., 2019) is a comprehensive, multidisciplinary resource aimed at quantitative evidence synthesis, but the initial chapters that focus on general topics and search guidance are applicable to both quantitative and qualitative research. Additionally, both the Campbell Collaboration (Campbell Collaboration, 2020), which focuses on social science research, and JBI Collaboration (Aromataris & Munn, 2020), an international healthcare research group, include guidance for qualitative evidence synthesis in their resources. We provided several examples of additional training opportunities in the literature review section of this article.

To make the most of any educational opportunity, it is important for researchers conducting evidence synthesis projects to have a thorough understanding of the vocabulary surrounding the method. The aforementioned PRISMA-P (Shamseer et al., 2015) and PRISMA-S (Rethlefsen et al., 2021) extensions both provide helpful glossaries for building a

TABLE 4 Explanation of selected search-related terms to SLRs and MAs.

Search-related term	Explanation	Examples
Database	Database that contains abstract and indexing information (e.g., title, abstract, authors, key terms) and/or the full text from <i>many</i> publishers; may be multidisciplinary or centered around a single subject/limited set of subject areas; may offer advanced search functionality (e.g., controlled vocabulary, Boolean operators, truncation)	ERIC, Compendex, PsycInfo, Inspec, Scopus, Social Sciences Citation Index, Conference Proceedings Citation Index, Academic Search Complete
Journal/publisher database	Database that contains the full text literature from a single publisher, or a limited number of publishers; typically the search functionality is limited (e.g., basic search with key terms)	ScienceDirect, IEEEExplore, Emerald Insight, Sage Journals, Wiley Online Library, ACM Digital Library
Vendor platform	Search engine that has the ability to search across multiple databases; different libraries may subscribe to different combinations of databases through a given platform; a database may be available on multiple platforms; search capabilities vary between platforms	Web of Science, Engineering Village, Ovid, ProQuest, EBSCO

foundation. Additionally, our results show a lack of adequate search reporting, which is consistent with previous studies, especially regarding the information sources searched in SLRs and MAs (Borrego et al., 2015; Harari et al., 2020). For this reason, we created Table 4 to further explain selected search-related terminology.

6.1.2 | Need for more librarian/expert search involvement

As noted by Borrego et al. (2014), it is crucial for engineering educators conducting SLRs to consult with a librarian or other search expert on selecting appropriate databases to search and creating search retrieval strategies. A librarian's role can range from a co-author with a significant focus on the methods section, such as Margaret Foster in Borrego et al. (2015), to a full project consultant where a librarian provides guidance on multiple aspects of gathering, analyzing, and reporting data, or a one-time consultant where a librarian may help create or peer-review a search strategy (McGowan et al., 2016). It is best practice for EE researchers to report the level of librarian involvement in the methods section of an SLR/MA and give proper acknowledgments for librarian contributions.

It is important to highlight there are different levels of librarian support efforts, as a librarian's capacity to collaborate with researchers on SLR/MA will vary from institution to institution. This could be particularly true for researchers at institutions where research is not a primary focus and could also vary by country, where librarians potentially have different roles and expectations. Further, not all librarians are trained in providing full methodological support to teams working on an SLR/MA. In these cases, we recommend researchers contact their institution's library at the start of the project to receive basic guidance in identifying available and relevant databases to search, how to list the databases and platforms searched, and to inquire what level of support might be available for building or reviewing the database search strategy. In cases where researchers do not have access to database experts to develop the search strategy, most librarians should be able to help researchers report their searches in a way that makes them reproducible, but those conversations would need to occur both before and after the search is run, rather than only after the fact. By having these conversations at the beginning of the process, researchers would know what details to capture and thus be able to report in their methods section. For SLRs/MAs, the search process is the method of creating the dataset used in that study and, for transparency, the methods section needs to include enough detail to make the study reproducible.

6.1.3 | Need for advocacy to publishers

To strengthen the quality of SLRs/MAs in the field of EE, there is a need to advocate for publishers to require authors to follow key reporting documents (e.g., PRISMA, PRISMA extensions) for submissions. Additionally, publishers should consider requesting librarians or other search experts to peer-review the methods sections of SLR submissions,

using standard reporting checklists. This could mirror practices of journals that ask initial reviewers whether the article requires a separate methodological review. EE publishers could recruit librarian reviewers from relevant professional societies, such as ASEE—Engineering Librarian Division (ELD) and the Association of College and Research Libraries (ACRL) Science and Technology Section (STS).

6.2 | Limitations

Our scoping review has a few limitations. First, we searched for studies where the authors used the terms “systematic review,” “systematic literature review,” “meta analysis,” or “meta-analysis” in the title, abstract, or subject, which potentially left out relevant EE studies that did not use these terminologies. Also, we used our best judgment to classify each article by a primary engineering discipline, but many articles in the dataset were multidisciplinary in nature. Lastly, we limited our search to articles published only in the English language.

6.3 | Suggestions for future research

The EE-specific guidance for SLRs created by Borrego et al. (2014) was a collaboration between engineering educators and a librarian skilled in SLR methods. Since this time, PRISMA has produced an updated PRISMA statement (Page et al., 2021) and several extensions designed to better clarify the reporting standard sections to assist authors with reporting their methods (Moher et al., 2015; Shamseer et al., 2015; Tricco et al., 2018), and the use of the SLR method has increased globally in the field of EE. For these reasons, we suggest it is time for the EE-specific guidance for evidence synthesis projects to be updated, ideally as a joint effort between EE researchers and STEM librarians skilled in evidence synthesis methods.

7 | CONCLUSION

We conducted a scoping review of engineering-related SLRs and MAs. This article shares the results of a sub-analysis of our data focused on EE. Our sub-analysis found that SLRs and MAs are increasing as published methodologies in EE, with 257 of the 276 discovered studies being published in 2015 or later. However, adherence to the best practices for reporting is not being followed universally, or in many cases by most of published articles. Given the role and credence lent to SLRs and MAs in evidence-based decision making, a failure to use the best practices can result in presentation of an incorrect summary due to missed evidence. The inclusion of librarians or research database experts trained in conducting SLRs can help increase the methodological quality and reporting of the methods. There is also an opportunity for EE journals and conferences to recruit librarians and other research database experts trained in conducting SLRs as peer reviewers to participate in evaluating the methods sections of submitted reviews. This need is particularly noteworthy given the increasing popularity of evidence synthesis projects within EE.

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DATA AVAILABILITY STATEMENT

Our data is available at <https://osf.io/de3mq/>.

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